U-TV C60 | Global parameter-map multiplier sweep

CIGS 0279 / One fixed noisy 20% Hadamard acquisition / 100 multipliers / 512 PDHG iterations

Experiment design

Image: CIGS 0279 (test permutation position 916/1000), 256 x 256 pixels.

Forward model: normalised J=8 Hadamard; 20% multilevel sampling

with coarse J=6 (13,107 measurements). Noise: fixed NPL-default.

C60: epoch 60; U-Net 3 blocks, 56 filters; CG input (20 iterations).

Predict the positive parameter map once and keep it frozen.

Run 512 MRpro-PDHG iterations for each rho = 0.1, 0.2, ..., 10.0.

Start every reconstruction from the same zero image.

$$\widehat{x}_\rho \approx \arg\min_x \left[\tfrac{1}{2}\|Ax-y\|_2^2 + \rho\,\mathrm{TV}_{\Lambda_{\mathrm{C60}}}(x)\right]$$

$$\Lambda_\rho = \rho\,\Lambda_{\mathrm{C60}}, \qquad \rho^* = \arg\max_{\rho\in\{0.1,\ldots,10.0\}} \mathrm{PSNR}(\widehat{x}_\rho, x_{\mathrm{GT}})$$

Quantitative results

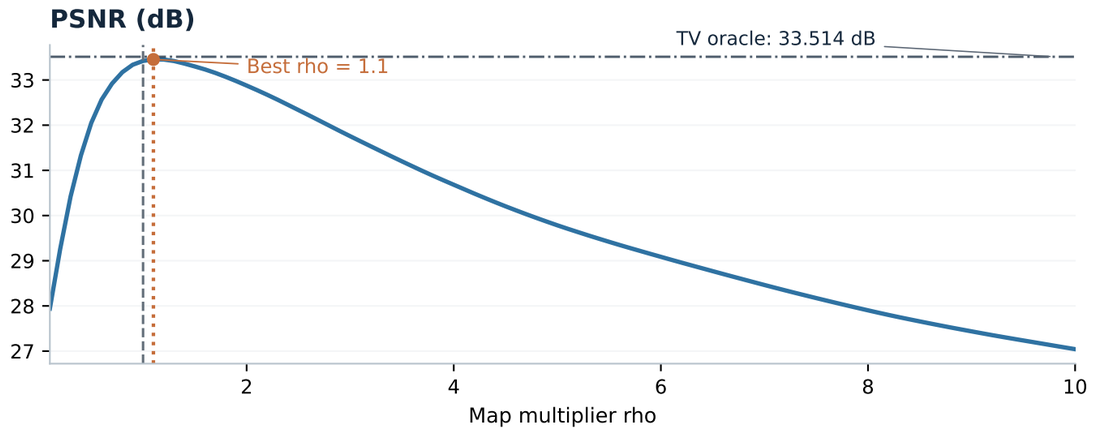
Method	PSNR (dB)	SSIM	HaarPSI	MSE
Zero-filled	26.513	0.5039	0.6453	2.232e-03
TV oracle, 512 iters	33.514	0.9173	0.8294	4.453e-04
C60, rho=1	33.422	0.9372	0.8287	4.548e-04
C60, rho=1.1	**33.454**	**0.9357**	**0.8208**	**4.515e-04**

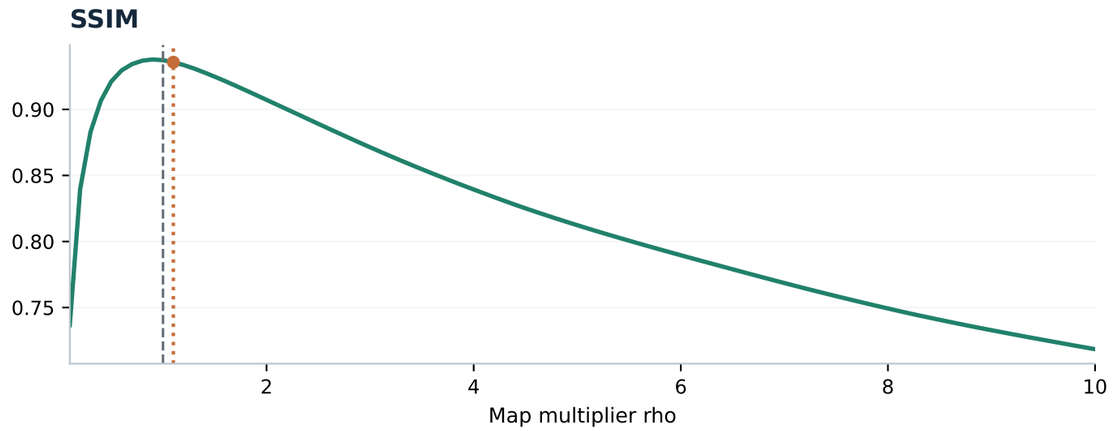
Best PSNR: $\rho^* = 1.1$ | **Gain over C60: +0.032 dB**

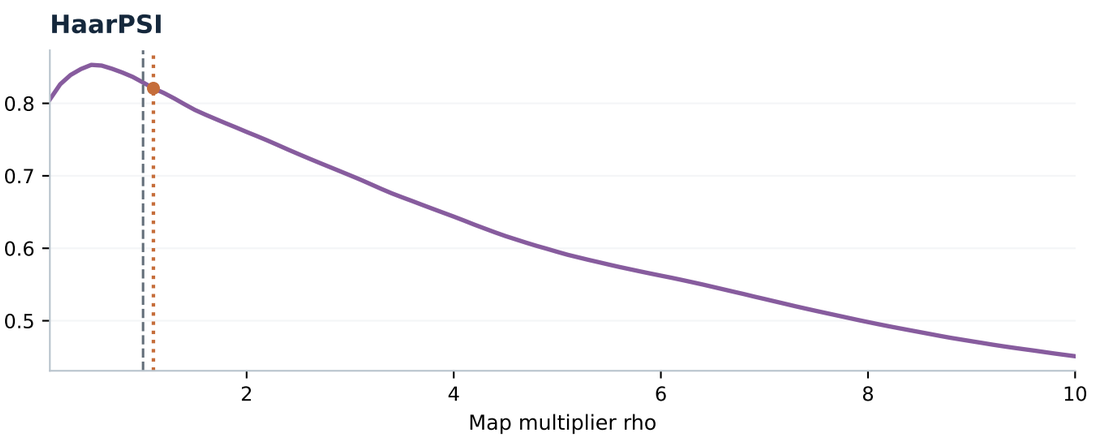
TV oracle remains +0.060 dB higher in PSNR than tuned C60.

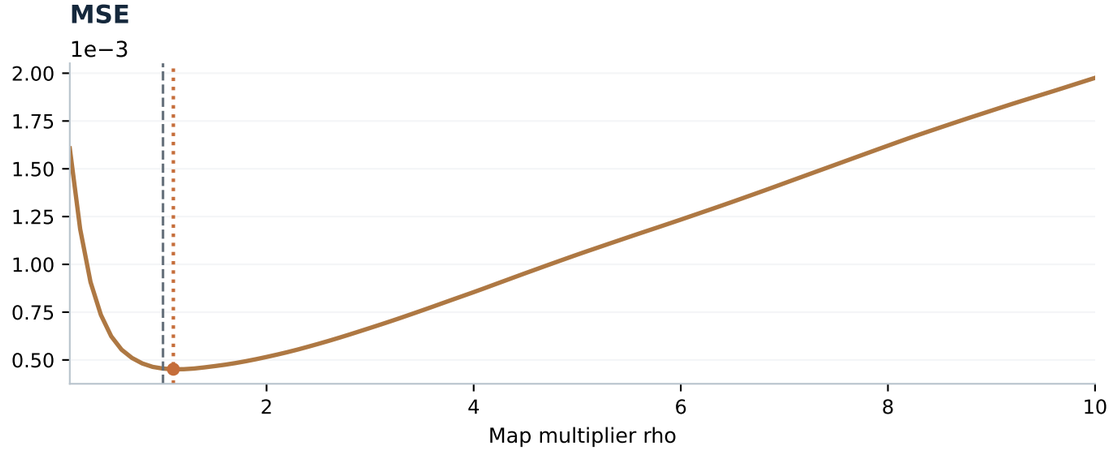
At rho=1.1, SSIM and HaarPSI both decrease relative to rho=1.

TV oracle lambda = 0.0322552321 (saved, 512 iterations).









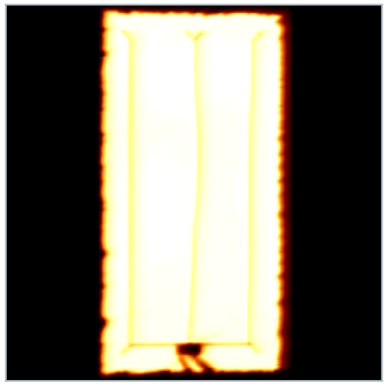
Per-image oracle analysis on one held-out test image. Not evidence of generalisation.

Reconstruction and parameter-map comparison

All images use the same frozen observation and the AFM colour scale [0, 1].

Ground truth, baselines, and the PSNR-selected multiplier

Ground truth

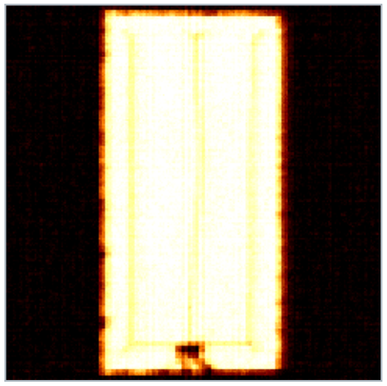


PSNR: inf

SSIM 1.0000 | HaarPSI 1.0000

MSE 0.000e+00

Zero-filled

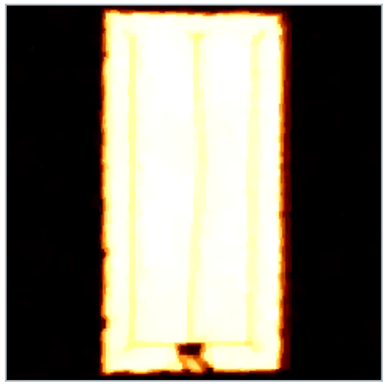


PSNR: 26.513 dB

SSIM 0.5039 | HaarPSI 0.6453

MSE 2.232e-03

Scalar TV oracle (512)

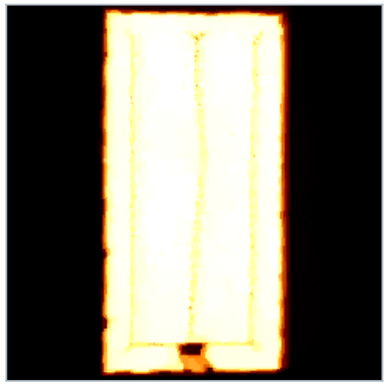


PSNR: 33.514 dB

SSIM 0.9173 | HaarPSI 0.8294

MSE 4.453e-04

C60, rho = 1

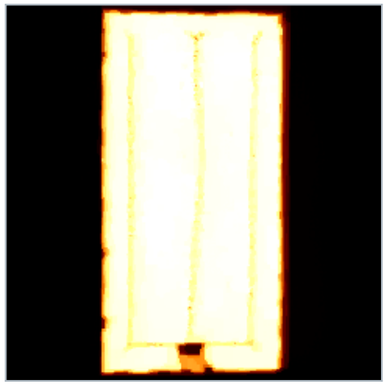


PSNR: 33.422 dB

SSIM 0.9372 | HaarPSI 0.8287

MSE 4.548e-04

C60, rho = 1.1



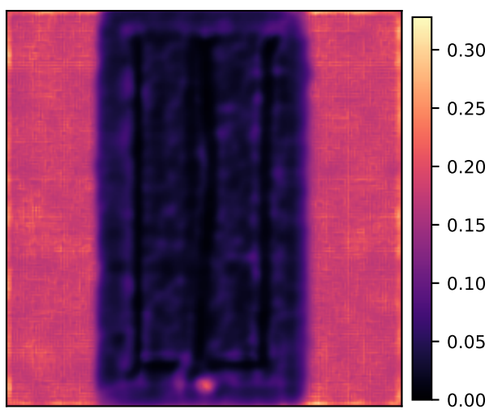
PSNR: 33.454 dB

SSIM 0.9357 | HaarPSI 0.8208

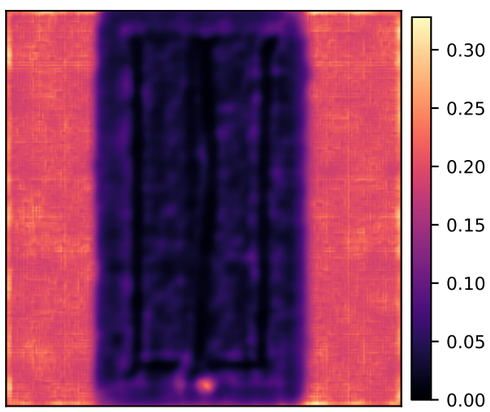
MSE 4.515e-04

Effect of rho on the learned regularisation map

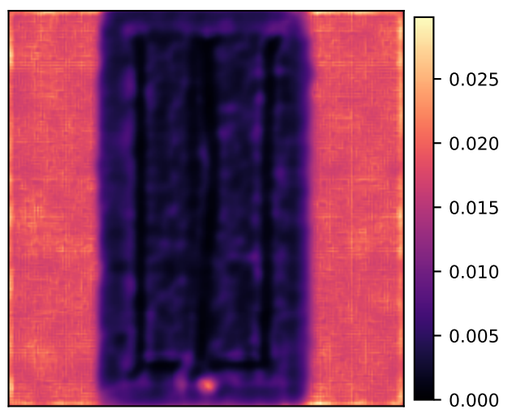
Original Λ_{C60}



Scaled map, $\rho = 1.1$



Difference, $\Delta\Lambda = 0.1\Lambda$



Learned map: min 0.0044, median 0.0814, max 0.2981.

The scalar changes the strength everywhere; it does not change the relative spatial pattern.